Resolvent, Scattering, and Heat Dynamics of a One-Dimensional Delta Interaction

Route-A source-local certificate HCS-C288

2 September 2026

Abstract

We give a convention-complete solution of the one-center delta interaction on the real line in units $\hbar = 2m = 1$. The closed quadratic form yields the exact continuity and derivative-jump domain, and a scalar interface equation gives the rank-one resolvent.

1 The frozen singular Hamiltonian

For $\alpha \in \mathbb{R}$, consider the form

$$q_\alpha[\psi] = \int_{\mathbb{R}} |\psi'(x)|^2 \, \mathrm{d}x + \alpha|\psi(0)|^2, \qquad \mathcal{D}(q_\alpha) = H^1(\mathbb{R}). \tag{1}$$

The one-dimensional trace inequality makes point evaluation infinitesimally form bounded relative to the Dirichlet integral. Thus (1) is closed and lower bounded and defines one self-adjoint operator H_α. Integration by parts identifies its domain:

$$\begin{aligned} \mathcal{D}(H_\alpha) = \{\psi \in H^2(\mathbb{R} \setminus \{0\}) : \psi(0+) &= \psi(0-), \\ \psi'(0+) - \psi'(0-) &= \alpha\psi(0)\}, \end{aligned} \tag{2}$$

and $H_\alpha \psi = -\psi''$ off zero. This normalization is the owner; changing the jump sign exchanges attractive and repulsive chambers.

Point interactions and their complete solvability are classical. The monograph of Albeverio, Gesztesy, Høegh-Krohn, and Holden contains a direct chapter on the one-center one-dimensional delta interaction [1]. Our contribution is the self-contained convention and executable closure below, not a priority claim.

2 One scalar equation gives the resolvent

For $\kappa > 0$, the free kernel at energy $-\kappa^2$ is

$$G_\kappa(x, y) = \frac{e^{-\kappa|x-y|}}{2\kappa}.$$

Theorem 1 (Rank-one resolvent)**.** *If $2\kappa + \alpha \neq 0$, then*

$$(H_\alpha + \kappa^2)^{-1}(x, y) = \frac{1}{2\kappa} \left\{ e^{-\kappa|x-y|} - \frac{\alpha}{2\kappa + \alpha} e^{-\kappa(|x|+|y|)} \right\}. \tag{3}$$

Proof. Away from $x = y, 0$, both exponentials solve the homogeneous equation. The free term is continuous and its derivative has jump -1 at $x = y$. At zero, the kernel value in (3) is $e^{-\kappa|y|}/(2\kappa+\alpha)$. The image term has derivative jump $\alpha e^{-\kappa|y|}/(2\kappa + \alpha)$, exactly α times that value. It therefore satisfies (2) and the resolvent source equation. Uniqueness follows from invertibility away from the displayed pole. □

The formula is a Krein rank-one correction. The exceptional equation $2\kappa + \alpha = 0$ has a positive solution precisely when $\alpha < 0$; it is spectral data, not a missing regular grid point.

References

[1] S. Albeverio, F. Gesztesy, R. Høegh-Krohn, and H. Holden, *Solvable Models in Quantum Mechanics*, Theoretical and Mathematical Physics, Springer, 1988, doi:10.1007/978-3-642-88201-2.